



DYNAMIC BEHAVIOUR

Software Design

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1. Design

1.1. Events

When operating a microwave there are multiple events that occur. These events will make the system switch between states. Regarding to the states, there are three states in which the microwave switches. The states are Idle, Running and Open Door. The events that occur, to switch between the states, are DoorClosed, ClockDone, DoorOpened.

1.2. State Diagram

As mentioned in 1.1. Events, there are three states. Within these states, there are multiple actions that occur; for example when it is in the Open Door state, the light is turned on and when it this state occurred while it was in a running state, the motor will be turned off. How these events and actions occur regarding to its state is displayed in Figure 1. State diagram microwave.

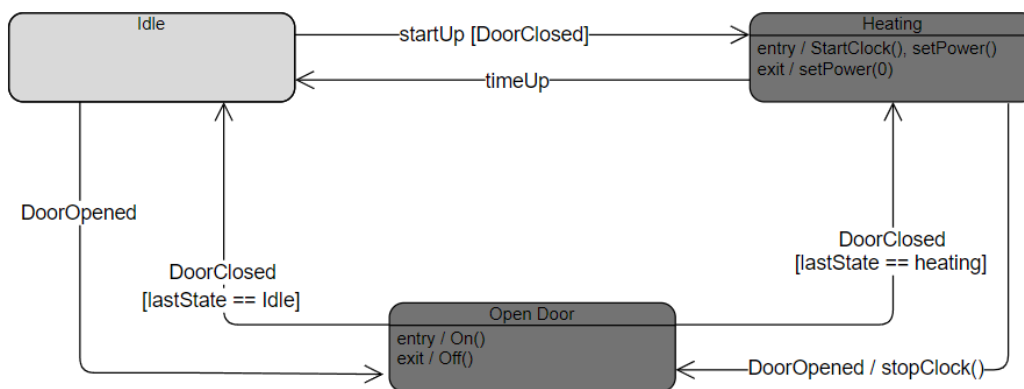


Figure 1. State diagram microwave

2. Code Testing

The approach of testing the code is based on google tests. These google tests are specified for each state switch. When a state switch occurs, there will be checked if all the called for actions occur and if they occur in the right order. The results of these tests are as following, see Figure 2. Results Google tests.

```
student@VirtualFontys:~/Desktop/Microwave/Microwave$ make test
Running main() from gmock_main.cc
[=====] Running 6 tests from 3 test cases.
[-----] Global test environment set-up.
[-----] 2 tests from TestOpenState
[ RUN    ] TestOpenState.test_close_door_event_back_to_heating
[ OK     ] TestOpenState.test_close_door_event_back_to_heating (0 ms)
[ RUN    ] TestOpenState.test_close_door_event_back_to_idle
[ OK     ] TestOpenState.test_close_door_event_back_to_idle (0 ms)
[-----] 2 tests from TestOpenState (0 ms total)
[-----] 2 tests from TestIdleState
[ RUN    ] TestIdleState.test_start_event
[ OK     ] TestIdleState.test_start_event (0 ms)
[ RUN    ] TestIdleState.test_open_door_event
[ OK     ] TestIdleState.test_open_door_event (0 ms)
[-----] 2 tests from TestIdleState (0 ms total)
[-----] 2 tests from TestHeatingState
[ RUN    ] TestHeatingState.test_time_up_event
[ OK     ] TestHeatingState.test_time_up_event (0 ms)
[ RUN    ] TestHeatingState.test_open_door_event
[ OK     ] TestHeatingState.test_open_door_event (0 ms)
[-----] 2 tests from TestHeatingState (0 ms total)
[-----] Global test environment tear-down
[=====] 6 tests from 3 test cases ran. (0 ms total)
[ PASSED ] 6 tests.
```

Figure 2. Results Google tests

3. Known Problems

The implementation of the code is only mocked, and therefore all the functions do not have the actual functions of what should be happening in real life. For example, there is a function called `light.On()`, but nothing will actually happen since the function does not include any written code.

Besides this, a known issue is that the `setPower` value is hardcoded within the program itself since the program ran into problems when wanting to reuse the `GetRequestedPower`, since there is nothing actually assigned to the power.